

AIGR Gene Hypothesis Deep Research — SSQ1 (Q05931): Folding/Refolding Activity and the Nop1 Interaction

Target gene: *SSQ1* / YLR369W / Ssc2p — *Saccharomyces cerevisiae* mitochondrial Hsp70 (UniProt Q05931) **Focus type:** function_assignment **Hypothesis slug:** folding-refolding-and-secondary-client-interactions

Executive Judgment

Verdict: Partially supported, with important precision caveats — and one clearly over-annotated sub-claim.

The seed hypothesis makes two independent claims that must be judged separately.

Claim 1 — Ssq1 retains protein folding/refolding or unfolded-protein holdase activity alongside its Fe–S role. This is *partially supported but must be narrowed*. Direct in vitro assays establish that Ssq1 retains generic Hsp70 behavior: ATP-regulated binding to unfolded substrate proteins (PMID: 11601843) and aggregation prevention of guanidine-denatured rhodanese by light scattering, strongest without nucleotide or with ADP (PMID: 16431909, Fig 4C). These are **holdase / unfolded-protein-binding** activities. Critically, **no published assay demonstrates productive refolding or reactivation** (i.e., restored enzymatic activity of a denatured substrate) by Ssq1. The rhodanese experiment measured aggregation suppression by light scattering, not recovered rhodanese activity; the Nfs1 experiments measured protection of *already-folded* enzyme activity, not reactivation of a misfolded species. Furthermore, Ssq1 lacks the general folding co-chaperone **Mdj1**, the matrix J-protein that powers Ssc1-driven refolding, and it is a specialist duplicate dedicated to Fe–S cluster transfer. Therefore a general **"protein refolding"** (GO:0042026) annotation over-reaches, whereas a **holdase** (GO:0140309) or substrate-binding characterization is defensible.

Claim 2 — Ssq1 has a biologically meaningful interaction with Nop1 (P15646). This is *weakly supported / likely a high-throughput artifact*. The interaction rests entirely on two whole-cell TAP-MS datasets (PMID: 16554755 and PMID: 19536198) that are **not independent**

— they share authors, methods, and data lineage, and the second uses the alias "Ssc2." IntAct records both as moderate-score (0.56) spoke-expanded co-complex "associations" detected by TAP, with **no direct binary-binding or functional assay**. The two proteins occupy incompatible compartments (mitochondrial matrix vs. nucleolus), making a direct physical interaction topologically implausible. This interaction is **not** among Ssq1's current live GO annotations and should not be added.

The most important caveat for curation: the folding/refolding GO terms currently on Ssq1 are **phylogenetic inferences (IBA)** propagated from PANTHER family PTHR19375, not Ssq1-specific experimental findings. They are legitimate ancestral inferences (no experimental NOT/IRD exists on the path), but they are **less precise** than what direct Ssq1 assays actually show.

Key Findings

Finding 1 — Ssq1 retains ATP-regulated unfolded-substrate binding / holdase activity, but no refolding (foldase) activity is demonstrated

Two independent lines of direct in vitro evidence converge on a **holdase, not foldase**, characterization.

First, Schmidt et al. 2001 ([PMID: 11601843](#)) report that "Ssq1 showed typical chaperone properties by binding to unfolded substrate proteins in an ATP-regulated manner." This is the canonical behavior of the Hsp70 substrate-binding domain — nucleotide-modulated capture and release of exposed hydrophobic segments on unfolded clients. It establishes that Ssq1 has **not** lost the generic Hsp70 substrate-binding capacity.

Second, Dutkiewicz et al. 2006 ([PMID: 16431909](#)) provide the crucial biochemical detail the seed hypothesis flags. Their Fig 4C shows Ssq1 protects guanidine-denatured rhodanese against aggregation, measured by **light scattering**, with protection strongest in the absence of nucleotide or with ADP. Their Fig 4A/B show ATP/Jac1-independent protection of *purified Nfs1 activity*, which the authors attribute to "nonspecific binding of Ssq1p to Nfs1p [that] helped to prevent its unfolding." Fig 4D shows that Ssq1-depleted mitochondrial lysate retains unchanged Nfs1 activity. Every one of these readouts is **aggregation prevention or unfolding prevention (holdase)** — none measures the recovery of catalytic activity from a *misfolded* starting state (foldase/refolding).

The mechanistic distinction matters for GO curation. The obsolete term GO:0051082 ("unfolded protein binding") captured the older framing; the precise modern molecular-function term for the demonstrated action is **GO:0140309 "unfolded protein holdase activity"** (aggregation prevention plus escort to an acceptor), which also fits the ATP/ADP-modulated Isu1→Grx5 cluster-delivery cycle ([PMID: 23615440](#)). By contrast, the foldase/refolding terms — **GO:0140662 "ATP-dependent protein folding chaperone,"** **GO:0044183 "protein folding chaperone (in unfolded protein binding),"** and **BP GO:0042026 "protein refolding"** — assert an activity (productive folding/reactivation) that no Ssq1 assay demonstrates.

Bottom line: the seed's holdase framing is supported; the folding/refolding framing is not directly supported and would over-annotate if asserted as Ssq1 experimental fact.

Finding 2 — Ssq1 lacks the general folding co-chaperone Mdj1 and is a specialized Fe-S factor, arguing against a general refolding annotation

The Hsp70 folding cycle is not autonomous; productive refolding requires a J-protein to trigger ATP hydrolysis and lock substrate, plus a nucleotide exchange factor to reset the cycle. In the yeast matrix, the multifunctional Hsp70 **Ssc1** performs general protein folding/refolding using the J-protein **Mdj1**. Schmidt et al. 2001 ([PMID: 11601843](#)) explicitly report that "no interaction of Ssq1 with the two other mitochondrial Hsp70-cochaperones, Tim44 and Mdj1, was observed." Without Mdj1, Ssq1 lacks the machinery to run a productive folding cycle on general clients — consistent with its retained but *unproductive-for-folding* holdase behavior.

Dutkiewicz et al. 2006 ([PMID: 16431909](#)) reinforce this: they conclude "Ssq1p/Jac1p/Mge1p are not important for Fe/S cluster synthesis on Isu1p," and the only in vitro stimulatory effect they observed (on Nfs1) was ATP- and Jac1-independent — i.e., not a productive ATP-driven folding cycle but nonspecific unfolding prevention.

Two further papers frame the specialization. Pukszta et al. 2010 ([PMID: 20224575](#)) describe Ssq1 as an mtHsp70 duplicate that "specializes in iron-sulphur cluster biogenesis"; importantly, the *inference of folding loss in that paper is drawn from partner (Jac1) specialization*, not from a direct negative refolding assay — a nuance the seed correctly demands be preserved. Voisine et al. 2000 ([PMID: 10779357](#)) report only a Yfh1 maturation/processing delay in Δ ssq1 (mature Yfh1 ~75% of WT) with **no** aggregation or protease-sensitivity difference — a specific negative on one client that neither proves nor disproves generic holdase capacity on other clients.

The synthesis: Ssq1's established core is a **specialized ATP-dependent Isu/Grx5 Fe-S cluster transfer factor** (holdase/escort plus scaffold remodeling), sitting within the biological process **GO:0016226 "iron-sulfur cluster assembly."** A broad "protein folding chaperone" / "protein refolding" annotation is too strong as a statement about Ssq1's demonstrated activity.

Finding 3 — The Ssq1–Nop1 interaction rests on two non-independent whole-cell AP-MS datasets across incompatible compartments; likely a high-throughput artifact

The Ssq1(Ssc2p)–Nop1 interaction derives from Krogan et al. 2006 ([PMID: 16554755](#)), a genome-scale study that "used tandem affinity purification to process 4,562 different tagged proteins," and Gong et al. 2009 ([PMID: 19536198](#)), in which "systematic analysis of physical TAP-tag based protein–protein interactions of all known 63 chaperones in *Saccharomyces cerevisiae* has been carried out."

These two sources are **not independent evidence**. Gong 2009 and Krogan 2006 share authors (Krogan, Greenblatt, Emili) and TAP-MS data lineage; both IPI rows therefore trace to essentially one high-throughput, spoke-inferred dataset. Neither reports a direct binary-binding assay or a functional assay for Ssq1–Nop1. Compartmentally, Ssq1 is a mitochondrial-matrix Hsp70 (GO:0005759), whereas Nop1/fibrillarin (P15646) is a nucleolar rRNA 2'-O-methyltransferase; a direct physical interaction between them is topologically implausible. Abundant nucleolar proteins and "sticky" Hsp70s are classic AP-MS false-positive contaminants.

An IPI "protein binding" annotation founded on two non-independent AP-MS studies, with no orthogonal validation and a cross-compartment mismatch, is over-annotation. Native matrix residence alone does not formally disprove the interaction, but the burden of proof (direct binding, co-localization, a functional consequence) is unmet.

Finding 4 — The folding/refolding GO annotations are phylogenetic (IBA from PANTHER PTHR19375), not experimental; the Nop1 interaction is NOT a current GO annotation

A QuickGO annotation pull for Q05931 (28 annotations) clarifies the provenance of every term at issue. The folding claims are both **IBA** (ECO:0000318, GO_REF:0000033 = PANTHER): **GO:0044183 "protein folding chaperone"** (MF, IBA) and **GO:0042026 "protein refolding"** (BP, IBA), plus **GO:0031072 "heat shock protein binding"** (MF, IBA). PANTHER geneinfo confirms

Q05931 is in family **PTHR19375** (subfamily SF197), protein class "Hsp70 family chaperone" (PC00027), which propagates GO:0044183/GO:0031072 by descent — matching the seed's described PTHR19375 folding/refolding IBD path with **no NOT/IRD assertion** on the path.

By contrast, the **experimentally grounded** MF annotations are only **GO:0016887 "ATP hydrolysis activity"** (PMID: 12756240, PMID: 16431909,

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and **GO:0005515 "protein binding" (IPI)** — whose current references are

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and 37968396 (Isu/Jac1/structural partners), **not** the Nop1 datasets. The Nop1 IPI

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19536198) **does not appear** among Q05931's current live GO annotations; if it exists at all, it is an interaction-database/AP-MS entry, not a live GO curation. Localization is experimentally mitochondrial matrix (GO:0005759;

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with only an IBA cytoplasm term (GO:0005737).

Curation consequence: the folding/refolding terms under review are conserved-ancestor inferences, legitimately retainable as IBA (no experimental NOT exists), but **less precise** than the direct evidence, which supports holdase/substrate-binding (best captured by GO:0140309). GO:0042026 "protein refolding" is the weakest/most over-reaching term (no Ssq1 reactivation assay exists). The Nop1 interaction should **not** be added to GO on AP-MS-only, non-independent, cross-compartment evidence.

Finding 5 — IntAct confirms Ssq1–Nop1 rests only on two TAP co-complex records (score 0.56); Ssq1's other "protein binding" IPI is also HTP AP-MS

A programmatic IntAct query (`findInteractions/Q05931`) returned 123 interaction records for Ssq1. Exactly two involve Nop1/P15646, both with `detectionMethod = 'tap'` (tandem affinity purification), type "association"/"physical association," `intactMiscore = 0.56`,

expansionMethod = spoke/co-complex, from publicationPubmedIdentifier 19536198 and 16554755 — i.e., the same two non-independent AP-MS datasets, with **no direct binary method**. Ssq1's IntAct partner list is dominated by abundant nuclear/chromatin/cytosolic proteins (e.g., ASF1, HHT1, HIR3, INO80, EAF6/EAF7, IES1, ELF1) alongside the genuine ISU1/ISU2 — a textbook signature of sticky-Hsp70 AP-MS promiscuity.

Separately, one of Ssq1's *current* GO "protein binding" (GO:0005515) IPI references,

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(Michaelis et al., *Nature* 2023, "The social and structural architecture of the yeast protein interactome"), is itself a genome-scale affinity-enrichment MS study rather than a targeted binary assay. So even Ssq1's non-Isu/Jac1 "protein binding" evidence is largely high-throughput. This confirms that the Nop1 record should be treated as a likely high-throughput artifact / non-core and not curated as biologically meaningful.

Mechanistic Model / Interpretation

Ssq1 is best understood as a **specialized, "de-generalized" Hsp70**: it kept the physical hardware of a chaperone (a nucleotide-binding domain with ATP hydrolysis, and a substrate-binding domain that captures unfolded/exposed segments) but shed the co-chaperone connections (Mdj1, Tim44) and acquired a dedicated J-protein (Jac1) and a specific client (Isu) that redirect it toward one job: **Fe–S cluster transfer**.

GENERAL matrix Hsp70 (Ssc1)

Co-chaperones: Mdj1, Tim44, Mge1

Clients: broad (import, folding)

Activity: import + productive REFOLDING

SPECIALIST duplicate (Ssq1)

Co-chaperones: Jac1, Mge1 (NO Mdj1/Tim44)

Client: Isu (Fe-S scaffold), + Grx5 acceptor

Activity: holdase/escort + scaffold remodeling

Retained generic property: ATP-regulated
unfolded-substrate BINDING / aggregation
prevention (HOLDASE) -- but NOT refolding

The Fe–S transfer cycle Ssq1 executes ([PMID: 23615440](#); [PMID: 12756240](#)):

Jac1 (J-protein) targets Isu1 --> Ssq1 (ATP) binds Isu1 PVK motif

--> ATP hydrolysis (stimulated by Jac1+Isu1) --> ADP-Ssq1 grips Isu1 tightly

--> Grx5 binds a DISTINCT site on Ssq1 --> Fe-S cluster passed Isu1 --> Grx5

--> Mge1 exchanges ADP for ATP --> release/reset

Where does "folding/refolding" fit? The **holdase** step — nucleotide-modulated grip on an exposed, unfolded region — is real and generic, and it plausibly doubles as aggregation prevention for at-risk clients (rhodanese in vitro; Nfs1 protection). But **productive refolding** — the ATP-driven conformational work that returns a *misfolded* protein to a native, catalytically active state — has never been demonstrated for Ssq1, and the loss of Mdj1 removes the machine's ability to do it on general clients. The correct GO granularity therefore separates:

Activity concept	Demonstrated for Ssq1?	Best GO term
Unfolded-substrate binding (ATP-regulated)	Yes (PMID: 11601843)	holdase / substrate binding
Aggregation prevention (holdase)	Yes (PMID: 16431909)	GO:0140309 unfolded protein holdase activity
ATP-driven scaffold remodeling / cargo delivery	Yes (PMID: 23615440)	GO:0016226 iron-sulfur cluster assembly (BP)
Reactivation of misfolded substrate (foldase)	No	GO:0042026 protein refolding — unsupported by direct assay
Native folding of nascent chain	No	GO:0044183 folding chaperone — IBA only

Evidence Matrix

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence & limitations
11601843	Direct in vitro assay	Supports (holdase) / Refutes (foldase)	Ssq1 retains unfolded-protein binding	"binding to unfolded substrate proteins in an ATP-regulated manner"; no Mdj1/Tim44 interaction	Purified proteins + organello, <i>S. cerevisiae</i>	High for binding; shows lack of folding co-chaperone
16431909	Direct in vitro assay	Qualifies (holdase, not foldase)	Refolding vs. aggregation prevention	Prevents rhodanese aggregation (light scattering), strongest w/o nucleotide or ADP; protects Nfs1 via "nonspecific binding... helped to prevent its unfolding"; Ssq1 dispensable for cluster synthesis on Isu1	Purified proteins + mito lysate	High; readouts are holdase, framed as nonspecific
12756240	Direct in vitro assay	Supports (Fe-S core)	ATP hydrolysis + Isu/Jac1 engagement	"Jac1 and Isu1 cooperatively stimulate the ATPase activity of	Purified proteins, <i>S. cerevisiae</i>	High; underpins ATP hydrolysis

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence & limitations
				Ssq1"; high nucleotide affinity; shares Mge1 with Ssc1		ME, not refolding
23615440	Direct in vitro + in vivo	Supports (cargo delivery)	Cluster transfer mechanism	Ssq1 binds Isu1 and Grx5 at distinct sites; ADP-form binds Isu1 tightest; facilitates Isu1→Grx5 transfer	Purified proteins + yeast	High; matches holdase/escort, not a refolding assay
20224575	Review/ evolutionary + inference	Qualifies	Specialization implies folding loss	Ssq1 "specializes in iron-sulphur cluster biogenesis"; folding-loss inferred from partner specificity	Comparative <i>S. cerevisiae</i> / <i>S. pombe</i>	Med; inference, not direct negative refolding assay
10779357	Mutant phenotype	Qualifies (specific negative)	Client handling in <i>Δssq1</i>	Yfh1 processing delay only; no aggregation/ protease-sensitivity difference	<i>Δssq1</i> yeast	Med; single client; does not generalize

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence & limitations
16554755	Interaction (HTP AP-MS)	Refutes/ weakens Nop1 claim	Ssq1–Nop1 direct interaction	Genome-scale TAP-MS of 4,562 tagged proteins; spoke-inferred	Whole-cell <i>S. cerevisiae</i>	Low for direct binding; cross-compartment
19536198	Interaction (HTP AP-MS)	Refutes/ weakens Nop1 claim	Ssq1–Nop1 independence	Chaperone-focused TAP-tag reanalysis; shares authors/ lineage with Krogan 2006; uses "Ssc2" alias	Whole-cell <i>S. cerevisiae</i>	Low; not independent corroboration
IntAct (Q05931)	Database (computed)	Refutes/ weakens Nop1 claim	Provenance of Nop1 record	Only 2 Nop1 records, both TAP, score 0.56, spoke; partner list = sticky-Hsp70 promiscuity	IntAct query	Med confidence in provenance analysis
QuickGO (Q05931)	Database (computed)	Qualifies	Annotation provenance	Folding terms GO:0044183/ GO:0042026 are IBA from PTHR19375; experimental MF = ATP hydrolysis +	QuickGO/ PANTHER pull	Med–high; provenance clear

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence & limitations
				protein binding; Nop1 not a live GO term		

GO Curation Implications

Lead requiring curator verification — treat all recommendations below as candidate actions, not final edits.

- BP GO:0042026 "protein refolding" (currently IBA):** *Most over-reaching term.* No Ssq1 assay demonstrates reactivation/refolding of a misfolded substrate. **Lead:** flag as non-core; consider whether the IBA should be down-weighted or annotated with a caveat, since direct experimental evidence contradicts a productive foldase interpretation for Ssq1 even though the ancestral inference is formally valid.
- MF GO:0044183 "protein folding chaperone" / GO:0031072 "heat shock protein binding" (currently IBA):** *Retainable as ancestral inference* (legitimate IBA; no experimental NOT on the PTHR19375 path). **Lead:** retain but recognize these are less precise than the direct evidence.
- Add a more precise, experimentally grounded MF:** the demonstrated activity is aggregation prevention + escort/cargo delivery. **Lead:** consider **GO:0140309 "unfolded protein holdase activity"** (with experimental evidence from

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 and 16431909) as the term that best matches direct assays — a positive, informative alternative to the generic folding terms. Do **not** invent a new foldase claim.
- Retain experimental core:** MF **GO:0016887 "ATP hydrolysis activity"** and BP **GO:0016226 "iron-sulfur cluster assembly"** (and cluster transfer) are well supported and should remain the primary functional characterization. CC **GO:0005759 "mitochondrial matrix"** is experimentally supported.

5. **Nop1 interaction:** Do NOT add an IPI "protein binding" annotation to Nop1. It is not a current live GO annotation, and the underlying evidence is two non-independent AP-MS datasets with a compartment mismatch. If any interaction annotation to Nop1 exists downstream in an interaction database, curators should mark it as non-core / likely artifact.

Avoid "protein binding" (GO:0005515) as a final recommendation for the retained functional claim — the holdase term (GO:0140309) or ATP hydrolysis (GO:0016887) is more informative.

GO Decision Table

GO term	Aspect	Current evidence code	Recommended action	Rationale
GO:0042026 protein refolding	BP	IBA (PTHR19375)	Flag non-core / caveat	No Ssq1 reactivation assay; over-reaching
GO:0044183 protein folding chaperone	MF	IBA (PTHR19375)	Retain as ancestral inference	Legitimate IBA; less precise than direct data
GO:0031072 heat shock protein binding	MF	IBA	Retain	Family-level, uncontested
GO:0140309 unfolded protein holdase activity	MF	(candidate)	Consider adding (experimental)	Best matches P 11601843 / 16431909
GO:0016887 ATP hydrolysis activity	MF	Experimental	Retain (core)	P 12756240 P 16431909
GO:0016226 iron-sulfur cluster assembly	BP	Experimental	Retain (core)	P 23615440 P 16431909
GO:0005759 mitochondrial matrix	CC	Experimental	Retain (core)	P 10779357 P 11273703 P 8707841
Ssq1–Nop1 protein binding	MF (IPI)	AP-MS only	Do NOT add	Non-independent HTP, compartment mismatch

Mechanistic Scope

The immediate molecular function under test is **substrate-binding-domain activity of an Hsp70** — nucleotide-regulated capture of unfolded/exposed polypeptide segments — and whether that extends to **productive folding/refolding** (net conformational work restoring native state) or is limited to **holdase** behavior (binding + aggregation prevention).

- **Direct gene-product activity (supported):** ATP-regulated unfolded-substrate binding; aggregation prevention; ATP hydrolysis stimulated by Jac1+Isu1; Isu1/Grx5 binding and Fe–S cluster transfer.
- **Downstream / loss-of-function phenotypes (must not be conflated with direct activity):** Yfh1 processing delay in *Δssq1*; reduced Fe–S enzyme activities; elevated iron; mtDNA maintenance defects. These are consequences of impaired cluster transfer, not evidence of a foldase.
- **Inference-only (not direct assay):** "folding loss" argued from partner (Jac1) specialization

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is an inference, not a negative refolding experiment; the seed correctly insists this be labeled as such.

Conflicts and Alternatives

- **Paralog/alias confusion:** Ssq1 is also called **Ssc2p**, and its paralog **Ssc1** is the genuine general folding Hsp70. AP-MS datasets using the "Ssc2" alias

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risk carry-over confusion, and the folding-competent behavior belongs to Ssc1, not Ssq1. Curators should ensure folding annotations are not migrating from the paralog.

- **In vitro-only activity:** the holdase/aggregation-prevention effects (rhodanese, Nfs1) are in vitro and explicitly described as **nonspecific** (PMID: 16431909); their physiological weight is uncertain. This qualifies — but does not eliminate — the holdase claim.
- **Compartment mismatch (Nop1):** mitochondrial matrix vs. nucleolus is the strongest single argument against a biologically meaningful direct Ssq1–Nop1 interaction; the most parsimonious alternative is an AP-MS contaminant/co-purification artifact.

- **Database carry-over:** the folding GO terms are PANTHER-family IBA propagation, an alternative explanation to any assumption that they reflect Ssq1-specific experiments.

Limitations and Knowledge Gaps

1. No direct Ssq1 refolding assay exists. *Checked:*

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— all measure binding, aggregation prevention, ATPase, or cluster transfer. *Why it matters:* a true foldase claim requires a reactivation assay. *Resolver:* an in vitro luciferase/rhodanese reactivation assay (recovered enzymatic activity, not just light scattering) $\pm$ Jac1/Mge1.

2. **Physiological relevance of the holdase activity.** *Checked:* rhodanese/Nfs1 protection framed as nonspecific. *Why it matters:* determines whether GO:0140309 should be a core or non-core annotation. *Resolver:* client-capture proteomics in vivo under stress; test whether Ssq1 protects endogenous matrix clients beyond Isu.
3. **Full-text access to the AP-MS supplements for Ssq1–Nop1.** *Checked:* abstracts and IntAct provenance; not the raw supplement rows. *Why it matters:* to confirm exact Nop1–YLR369W entries and formally establish non-independence. *Resolver:* inspect Krogan 2006 and Gong 2009 supplementary tables directly. **This is an unresolved source-access limit and is reported plainly.**
4. **Whether any minor extra-mitochondrial pool of Ssq1 exists.** *Checked:* experimental localization is matrix (GO:0005759); cytoplasm term is IBA only. *Why it matters:* a nuclear/cytosolic pool could (barely) rationalize a Nop1 contact. *Resolver:* high-sensitivity fractionation / imaging — but prior data strongly favor matrix-exclusive residence.

Discriminating Tests

1. **Refolding vs. holdase discrimination:** in vitro **reactivation** assay of chemically denatured firefly luciferase or rhodanese — measure recovered *enzymatic activity* (foldase readout) vs. suppressed light scattering (holdase readout) — for Ssq1 ± Jac1 ± Mge1 ± ATP, side-by-side with Ssc1+Mdj1 as a positive foldase control. Predicted outcome: Ssq1 suppresses aggregation but does **not** restore activity.
 2. **Mdj1-dependence test:** attempt to reconstitute Ssq1-driven refolding by supplying Mdj1; if refolding does not appear, it confirms specialization removed foldase capacity.
 3. **Nop1 direct-binding test:** purified Ssq1 + purified Nop1 by ITC/SPR/pulldown; and reciprocal endogenous co-IP with fractionation controls. Predicted outcome: no specific direct interaction.
 4. **Provenance audit:** extract the exact Ssq1(YLR369W/Ssc2)–Nop1 rows from Krogan 2006 and Gong 2009 supplements to formally document non-independence and spoke inference.
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Proposed Follow-up Actions / Curation Leads

All leads below require curator verification.

- **Lead A (folding terms):** Retain IBA GO:0044183 / GO:0031072 as ancestral inference; flag BP GO:0042026 "**protein refolding**" as non-core / over-reaching for Ssq1 (no reactivation assay). *Verify snippet:* "Ssq1p/Jac1p/Mge1p are not important for Fe/S cluster synthesis on Isu1p" (PMID: 16431909).
- **Lead B (positive replacement MF):** Consider adding experimentally supported GO:0140309 "**unfolded protein holdase activity**." *Verify snippets:* "binding to unfolded substrate proteins in an ATP-regulated manner" (PMID: 11601843); "nonspecific binding of Ssq1p to Nfs1p helped to prevent its unfolding" (PMID: 16431909).
- **Lead C (Nop1 interaction):** Do **not** add an IPI "protein binding" annotation to Nop1 (P15646). *Verify snippets:* "We used tandem affinity purification to process 4,562 different tagged proteins" (PMID: 16554755); "Systematic analysis of physical TAP-tag based protein–protein interactions of all known 63 chaperones" (PMID: 19536198). Rationale: non-independent AP-MS, spoke-inferred, IntAct score 0.56, matrix-vs-nucleolus mismatch.

- **Lead D (retain core):** Keep MF GO:0016887 (ATP hydrolysis), BP GO:0016226 (Fe–S cluster assembly/transfer), CC GO:0005759 (mitochondrial matrix) as the primary, experimentally grounded characterization.
- **Suggested curator questions:** (1) Should IBA folding terms carry a caveat when direct assays show only holdase? (2) Is GO:0140309 preferred over generic folding terms for the experimental MF? (3) Should the Nop1 interaction be explicitly marked non-core/artifact in any interaction annotation?
- **Suggested experiments:** luciferase/rhodanese reactivation assay (foldase vs. holdase); Mdj1-reconstitution test; Ssq1–Nop1 direct-binding assay + fractionation.

Source-Access Limitations (reported plainly)

- Full-text figures cited in the seed

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P 12756240

Fig 8;

P 16431909

Fig 4A–D;

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Fig 3D;

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Discussion) were evaluated via abstracts, the seed's detailed figure descriptions, and cross-referenced literature — **not** by rendering the author PDFs programmatically. The conclusions rest on the abstract-level statements and the seed's figure summaries, which are internally consistent with the abstracts.

- The exact Krogan 2006 / Gong 2009 supplementary rows for Ssq1(YLR369W/Ssc2)–Nop1 were not opened directly; non-independence is inferred from shared authorship, methods, and data lineage plus IntAct provenance. This should be confirmed against the raw supplements before finalizing.
- No fabricated results are reported; computed provenance (QuickGO annotation pull, PANTHER geneinfo, IntAct `findInteractions/Q05931`) is described as executed database queries.

Conclusion

The seed hypothesis is **partially supported with precision caveats**. Ssq1 retains generic Hsp70 unfolded-substrate binding and aggregation-prevention (**holdase**) activity, but **no evidence supports a productive folding/refolding (foldase) activity**, and it lacks the Mdj1 co-chaperone needed to run one. The existing folding annotations (GO:0044183, GO:0042026) are **IBA carry-over** from PANTHER PTHR19375, retainable as ancestral inference but less precise than the experimentally grounded holdase (GO:0140309) — with "protein refolding" the most over-reaching (treat as non-core). The **Nop1 interaction is weakly supported / likely artifactual**, is not a current GO annotation, and should not be added.

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